

Azure Administrator & Cloud Engineer

40-Day Live Instructor-Led Training Program

Welcome!

Thank you for choosing this Azure Administrator & Cloud Engineer program. We are excited to have you on this learning journey. This course is designed to help you move beyond theory and gain practical experience with real Azure administration tasks.

Over the next 40 days, you will progressively build your skills across Azure architecture, identity, security, storage, compute, networking, monitoring, automation, backup and disaster recovery. The goal is simple: by the end of the program, you should be able to confidently deploy, secure, manage, troubleshoot and automate Azure environments.

Thank you for investing your time in learning Azure. Stay consistent, complete the hands-on labs, ask questions, and practice the scenarios repeatedly. Your practical experience will be just as important as your understanding of the concepts.

Course Curriculum

The 40-day curriculum is organized into eight technical modules followed by a separate capstone project. Each module combines concepts, administration tasks, troubleshooting and hands-on practice.

Module	Days	Primary Focus
1	Days 1–5	Azure Fundamentals, Architecture & Governance
2	Days 6–9	Microsoft Entra ID, RBAC & Security
3	Days 10–14	Azure Storage
4	Days 15–20	Azure Compute & Application Services
5	Days 21–27	Azure Networking
6	Days 28–31	Azure Monitoring & Troubleshooting
7	Days 32–35	Azure Administration, CLI, PowerShell & Bicep
8	Days 36–40	Backup, Disaster Recovery & AZ-104 Preparation
Capstone	Final Project	End-to-End Azure Administration & Troubleshooting

Module 1: Azure Fundamentals, Architecture & Governance | Days 1–5

Build a strong foundation in Azure architecture and learn how Azure resources are organized, governed, managed and secured.

Topics

Cloud concepts; IaaS/PaaS/SaaS; Azure regions & Availability Zones; subscriptions; Management Groups; Resource Groups; resource providers; resource IDs; ARM; tags; Azure Policy; policy initiatives; policy effects; policy compliance & remediation; resource locks; Cost Management; cost analysis; budgets & alerts; Azure Advisor; Portal and Cloud Shell.

Hands-on

Set up an Azure environment; create and organize subscriptions/resource groups; configure resource providers; apply tagging standards; create and assign Azure Policies; review policy compliance; configure resource locks; analyze costs and create budgets.

Module 2: Microsoft Entra ID, RBAC & Security | Days 6–9

Learn how Azure manages identities and how administrators control access to resources and applications.

Topics

Microsoft Entra ID; users; groups; guest users; administrative units; RBAC and role scopes; built-in/custom roles; role assignments; inheritance; MFA; Conditional Access; SSPR; Managed Identities; Azure Key Vault; secrets; keys and certificates.

Hands-on

Create users and groups; configure RBAC at different scopes; test inherited permissions; implement MFA/Conditional Access and SSPR; create managed identities; securely access secrets using Key Vault; troubleshoot authorization failures.

Module 3: Azure Storage | Days 10–14

Learn to provision, secure and manage Azure storage for application data, files and objects.

Topics

Storage Accounts; Blob Storage; containers; access tiers; LRS/ZRS/GRS/GZRS; redundancy selection; lifecycle management; versioning; soft delete; snapshots; object replication; Azure Files; SMB; Azure File Sync basics; Microsoft Entra ID authorization; RBAC for Storage; SAS; access keys; encryption; storage firewalls; Private Endpoints; Storage Explorer; AzCopy.

Hands-on

Build storage solutions; create and manage blobs/files; configure Microsoft Entra/RBAC and SAS access; configure lifecycle and security policies; implement Private Endpoints; use Storage Explorer and AzCopy; recover deleted/versioned data.

Module 4: Azure Compute & Application Services | Days 15–20

Learn how to deploy, manage, scale and troubleshoot Azure workloads using VMs, VM Scale Sets, App Service and containers.

Topics

Windows/Linux VMs; VM sizing; disks; managed disks; disk snapshots; RDP/SSH; VM extensions; Availability Sets/Zones; VM Scale Sets and autoscaling; App Service; App Service Plans; Web Apps; application settings; environment variables; deployment slots & slot swapping; custom domains; TLS/SSL; scaling; backups; troubleshooting; Azure Container Registry; Container Instances; Container Apps.

Hands-on

Deploy Windows/Linux VMs; configure disks and extensions; connect using RDP/SSH/Bastion; configure availability and scaling; deploy Web Apps with staging/production slots; perform slot swaps and rollbacks; configure application settings and TLS; deploy container workloads; troubleshoot compute failures.

Module 5: Azure Networking | Days 21–27

Build secure and scalable Azure networks and understand how resources communicate with each other, the internet and on-premises environments.

Topics

VNets; subnets; IP addressing; public/private IPs; NICs; NSGs; ASGs; NSG rules and effective security rules; route tables/UDRs; system routes; VNet Peering; Service Endpoints; Private Endpoints; Private DNS; Azure Bastion; VPN Gateway; Azure DNS; Load Balancer; Application Gateway and WAF. Load Balancer: Public vs Internal; frontend IP configurations; backend pools; health probes; load-balancing rules; inbound NAT rules; HA ports. Application Gateway: frontend listeners; backend pools; HTTP settings; health probes; routing rules; TLS termination; WAF.

Hands-on

Build multi-subnet networks; configure public/private IP addressing; secure traffic with NSGs and ASGs; configure route tables and UDRs; implement VNet peering, Service Endpoints and Private Endpoints; configure Private DNS; deploy Bastion and VPN connectivity; build Load Balancer and Application Gateway solutions; troubleshoot connectivity and DNS failures.

Module 6: Azure Monitoring & Troubleshooting | Days 28–31

Learn how to monitor Azure environments, analyze resource health and logs, configure alerts and troubleshoot infrastructure and networking issues.

Topics

Azure Monitor; Metrics; Activity Logs; Resource Logs; Diagnostic Settings; Log Analytics; KQL; Alerts; Action Groups; Dashboards; Advisor; Service Health; Resource Health; Network Watcher. Troubleshooting: VM connectivity; NSG troubleshooting; effective routes; DNS resolution; Network Watcher connection troubleshoot; IP flow verify; packet capture; Log Analytics queries; resource health; platform issues.

Hands-on

Configure monitoring; enable diagnostic settings; send logs to Log Analytics; create alerts and action groups; analyze logs with KQL; troubleshoot VM and networking failures; investigate DNS and routing problems; use Network Watcher to identify connectivity issues.

Module 7: Azure Administration, CLI, PowerShell & Bicep | Days 32–35

Develop practical administration and automation skills using Azure CLI, PowerShell and Infrastructure as Code.

Topics

Cloud Shell; Azure CLI; PowerShell; resource/subscription management; resource groups; RBAC administration; VM and storage administration; ARM templates; Bicep; parameters; variables; modules; dependencies; deployment scopes; resource deployment.

Hands-on

Manage Azure resources from the command line; create and modify resources using Azure CLI and PowerShell; administer VMs/storage/RBAC; deploy resources using Bicep; use parameters and modules; troubleshoot failed deployments; deploy a complete Azure environment using Infrastructure as Code.

Module 8: Backup, Disaster Recovery & AZ-104 Preparation | Days 36–40

Learn to protect workloads and recover from failures, while preparing for integrated Azure administration scenarios.

Topics

Azure Backup; Recovery Services Vault; Backup Vault; backup policies; VM backup/restore; recovery points; Azure Site Recovery; replication; recovery plans; failover/failback; RPO; RTO; business continuity; integrated AZ-104 administration scenarios.

Hands-on

Configure VM backup and restore; create recovery points; configure replication; perform failover/failback exercises; review RPO/RTO; practice portal/CLI/PowerShell administration tasks; complete timed troubleshooting exercises; review common AZ-104 administration patterns.

Capstone Project: End-to-End Azure Administration

The capstone project is a separate practical project designed to bring together the skills developed throughout the course. You will work through a realistic Azure environment and demonstrate your ability to design, deploy, secure, monitor, troubleshoot, automate and recover workloads.

Capstone Objectives

- Secure a production-style VM environment using appropriate identity, RBAC, network and security controls.
- Build a secure storage solution with appropriate access, redundancy, lifecycle and recovery configuration.
- Deploy a highly available web application using appropriate compute and networking components.
- Configure monitoring, diagnostics, alerts and action groups.
- Troubleshoot intentional VM, networking, DNS, RBAC, storage and application failures.
- Automate the environment using Azure CLI, PowerShell and Bicep.
- Configure backup and disaster recovery and perform recovery exercises.
- Document the architecture, configuration decisions, troubleshooting steps and recovery process.

Scenario-Based Troubleshooting

- VM connectivity failure
- NSG blocking traffic
- Incorrect route or UDR configuration
- DNS resolution failure
- Private Endpoint connectivity failure
- RBAC authorization failure
- Storage access failure
- Failed application deployment

AZ-104 Exam Preparation

Complete integrated administration scenarios, practice portal/CLI/PowerShell-based tasks, review common troubleshooting patterns, perform timed hands-on exercises and validate the complete environment against AZ-104 administration objectives.

Learning Outcomes

- Understand Azure architecture, subscriptions, resource organization and governance.

- Implement identity, RBAC, Conditional Access, managed identities and Key Vault security.
- Design and secure Azure Storage using redundancy, access control, lifecycle and private networking.
- Deploy and administer VMs, VM Scale Sets, App Service and container workloads.
- Build secure Azure networks using VNets, NSGs, routing, private connectivity, VPN and application delivery services.
- Monitor Azure environments using Azure Monitor, Log Analytics, KQL, alerts and Network Watcher.
- Automate administration with Azure CLI, PowerShell and Bicep.
- Protect workloads with backup and disaster recovery and validate recovery procedures.
- Troubleshoot common Azure compute, networking, DNS, RBAC, storage and deployment failures.
- Demonstrate end-to-end Azure administration through the capstone project.